

From Generalization to Clinical Utility: A Strategic Alignment Pipeline for Qwen 3-4B in Mental-Health Dialogue

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Abstract

The rapid progress of large language models (LLMs) offers a path to scaling mental-health support, yet general-purpose models often fall short of clinical expectations in high-stakes dialogue. We study how to bridge this gap by aligning a lightweight open model, Qwen3-4B, toward clinically useful responses while maintaining reproducibility and evaluation rigor. Our framework proceeds in three steps. First, we run systematic prompt engineering, evaluating four prompt designs and selecting the strongest prompt as a fixed baseline. Second, we perform two-stage alignment: (i) QLoRA-based supervised fine-tuning on high-scoring mental-health responses, and (ii) Group Relative Policy Optimization (GRPO) to optimize model behavior against a multi-dimensional clinical rubric. Third, we evaluate with the Affective-Cognitive Agreement Framework, using an LLM jury and agreement-based analysis to reduce single-judge variance and improve reliability.

1 Introduction

Mental health care faces a persistent and widening global crisis. The World Health Organization reported that more than 1 billion people are living with mental health conditions, yet a large number lack access to mental health care and treatment (World Health Organization, 2025). LLMs have emerged as a promising avenue for expanding access to psychological support at scale, offering always-available, low-barrier psychological counseling.

Despite this promise, general-purpose LLMs remain unsuitable for clinical deployment. These models frequently produce responses that are generic, insufficiently actionable, or clinically unsafe when evaluated against professional standards. This inadequacy is most consequential in high-risk scenarios—such as active suicidal ideation or acute

crisis—where model outputs must meet strict standards of safety, empathy, and clinical appropriateness. The gap between what current LLMs offer and what mental health contexts demand motivates a need for domain-specific alignment.

Another concern is evaluation. Counseling-style dialogue is inherently multi-dimensional, spanning affective attunement, cognitive accuracy, actionability, and safety. Standard benchmarks designed for general NLP tasks fail to capture these clinically meaningful distinctions. Recent work has begun to address this gap: Badawi et al. (2025) introduce rubric-based scoring across seven clinically motivated attributes, organized into cognitive and affective dimensions, to assess when automated LLM judges can be trusted.

In this work, we present a systematic optimization pipeline targeting both alignment and evaluation for mental health dialogue. Our contributions are threefold:

- **Controlled baseline establishment.** We standardize generation settings for Qwen3-4B, including disabling thinking-mode output that consumes the token budget without improving clinical utility, providing a reproducible foundation for downstream comparisons.
- **Lightweight clinical alignment.** We apply a two-stage alignment pipeline: systematic prompt engineering to identify the most stable behavioral baseline, followed by parameter-efficient SFT via QLoRA and reinforcement learning from AI feedback (RLAIF) using GRPO to align model behavior with professional clinical standards.
- **Multi-judge evaluation.** To mitigate scoring bias inherent in single-model evaluation, we introduce an LLM Jury framework—a combined scoring system aggregating assessments from three distinct LLMs—providing more

robust and clinically grounded evaluation of generated responses.

Together, these components form an end-to-end pipeline that advances the clinical utility and evaluative rigor of LLM-based mental health dialogue systems. We demonstrate that targeted alignment, even at modest model scale, can meaningfully close the gap between general-purpose LLMs and professional counseling standards.

2 Related Work

Recent research on emotional support generation has identified structural challenges in alignment strategies. The DecoupledESC framework (Zhang et al., 2025) demonstrates that emotional strategy selection and surface realization are often entangled during training, and proposes a decoupled optimization approach to improve empathetic response quality. While this strategy-level disentanglement mitigates formulaic or “robotic” responses, it does not explicitly operationalize structured multi-dimensional clinical criteria, such as safety monitoring or actionable guidance, relevant to professional mental health contexts. Furthermore, Umucu et al. (2025) investigates optimization objectives targeting empathy in healthcare; however, its alignment primarily treats empathy as a unified objective and does not systematically examine potential trade-offs between cognitive support and affective resonance.

Evaluation reliability has emerged as another important concern. Chen et al. (2024) identify systematic biases in both human and LLM evaluators, including inflation effects on affective metrics and a “beauty bias” that favors linguistically polished responses over those that are clinically sound. Extending this concern to the evaluation paradigm, Wang et al. (2024) observe that relying on a single prompt can render LLMs as unfair evaluators. Moreover, because models exhibit high sensitivity to single-prompt configurations (Lior et al., 2025), we address this vulnerability by generating four variations to reliably select an optimal one. Since poorly prepared meta-prompts frequently lead to unsatisfactory outputs, we further explicitly optimized our instructions based on findings that increasing prompt length and lexical diversity (Wahle et al., 2024), alongside integrating detailed descriptions, context specifications, and step-by-step reasoning templates (Ye et al., 2024), significantly enhances prompt performance. Additionally, to

mitigate the intra-model bias inherent in relying on a single LLM as a judge (Zhao et al., 2025), we employ a collaborative panel of LLMs to jointly evaluate the responses.

On the data and benchmarking front, Chen et al. (2025) introduce a multi-turn counseling dataset emphasizing explainable reasoning processes. In parallel, Badawi et al. (2025) propose the MentalAlign-70k benchmark, a large-scale reliability framework designed to analyze LLM performance under high-risk scenarios using a dual-axis evaluation system. While these efforts substantially advance dataset construction and robustness analysis, current alignment pipelines rarely integrate structured clinical scores directly into training objectives. Furthermore, regarding the alignment optimization process, traditional reinforcement learning methods requiring additional value models often massively increase computational overhead. To address this, we adopt Group Relative Policy Optimization (GRPO) instead of Direct Preference Optimization (DPO), as Zhu et al. (2025) demonstrate that GRPO effectively lowers computational costs. Taken together, prior work has addressed strategy disentanglement, empathy-focused optimization, evaluator bias, and large-scale benchmarking. However, systematic multi-dimensional clinical alignment under rubric-driven optimization, coupled with reliability-aware evaluation protocols, remains comparatively underexplored.

3 Methodology

3.1 Dataset and Task Definition

The MentalBench-100K dataset is sourced from the MentalBench-Align repository on GitHub, focusing on 23 clinical conditions such as anxiety, depression, grief, and relationship issues. In MentalBench-100K, approximately 3% of the Qwen3 response field includes thinking content (e.g., <think> ...) mixed into the saved output. This “thinking” text appears to be treated as part of the model’s response and may therefore be included in subsequent evaluations in MentalAlign-70k. Additionally, 3% of the Qwen3 response fields are blank, likely due to the same issue.

To address these problems, we reconstructed our own baseline with 7 dimensions by disabling Qwen3’s thinking feature, ensuring that the thinking process neither appears in nor occupies the response content. The MentalAlign-70k dataset is built upon the first 1,000 conversations of

MentalBench-100K, where human experts and four LLM judges rated 10,000 responses across seven dimensions including Guidance and Informativeness. Our training set is selected from MentalAlign-70k by filtering for the two responses that achieved the highest average scores per conversation.

The input contains a system prompt and a user message. The system prompt defines the model’s “psychiatrist” persona and operating rules. The user message describes the individual’s mental health concerns. The output is the model’s text response, which should constitute a supportive and professional paragraph. Both input and output are plain text. Our data follows a single-turn (one question, one answer) dialogue format. From a research perspective, our goal is to improve the base model using QLoRA and Group Relative Policy Optimization (GRPO), and to evaluate outputs across 7 clinical dimensions using a Panel of LLM Judges (3 LLMs).

3.2 Data Preprocessing and Dataset Construction

We built a clean data pipeline to prepare inputs for fine-tuning and evaluation. We pulled raw data from the MentalBench-Align repository using two main files: `Human_judge.xlsx` (dimensional scores) and `MentalBench-100k.csv` (full conversational text).

3.2.1 Data Alignment and Text Cleaning

We stripped extra whitespace from the conversational contexts in both datasets. Using these cleaned contexts as keys, we mapped the unique Conversation IDs (ConvID) from the scoring dataset to the full 100K dialogue dataset, directly linking human ratings to the raw text.

3.2.2 Score Parsing and Formatting

The original scoring data came in wide format, with each model occupying its own column and scores embedded in text strings. We reshaped the table into long format using Pandas’ `melt` function, extracted numerical scores via regular expressions, mapped them to our seven evaluation dimensions (Guidance, Safety, Empathy, etc.), filled missing values with zeros, and converted all values to integers.

3.2.3 Building the Training Set

To create a high-quality dataset for LoRA fine-tuning, we filtered records carefully. We explicitly removed outputs generated by our baseline model

(Qwen3-4B) to prevent training on its own prior outputs. For the remaining models, we summed dimension scores to compute a `Total_Score` per response. We grouped data by ConvID, sorted by score, and retained only the top two highest-scoring responses per prompt. We then retrieved the corresponding response text, discarding any empty or malformed records. This produced a clean, high-quality training set (`dataset_sft`).

3.2.4 Building the Evaluation Set

To prevent data leakage, we constructed a test set entirely separate from the training data. We identified records in the 100K dataset lacking a ConvID—meaning they had never been seen or scored by human judges—and randomly sampled from these unseen queries to build our prompt engineering test set (`dataset_pe`). This ensures that future evaluations measure genuine generalization.

3.3 Model Selection

We selected Qwen3-4B (Qwen/Qwen3-4B) as our primary experimental model based on three key methodological considerations. First, utilizing it as the direct baseline model for the foundational dataset ensures a consistent underlying architecture, allowing us to construct a strictly controlled experimental environment that isolates extraneous variables and directly attributes performance improvements to our proposed optimization techniques. Second, the 4-billion-parameter scale is computationally feasible for our available resources, enabling efficient execution of subsequent workflows—such as applying QLoRA on the selected dataset—on a single local GPU with 24 GB VRAM or cloud-based T4 instances, ensuring rapid iteration and high reproducibility. Finally, the Qwen3 architecture demonstrates robust capabilities in instruction following and long-context processing, which are essential for capturing the nuanced sequential dynamics inherent in mental health counseling dialogues.

For the model evaluation pipeline, we adopt the LLM-as-a-Judge paradigm, establishing a review committee comprising three state-of-the-art models: GPT-5.4, Gemini 3 Flash, and Claude 4.5 Haiku. Given the high complexity and subjectivity of mental health dialogues, traditional n -gram-based metrics such as BLEU and ROUGE are insufficient for measuring empathy, safety, and coherence. We selected these models primarily for their advanced reasoning capabilities, which

demonstrate strong alignment with human expert judgments. To mitigate inherent biases from single-model evaluation, we tasked this committee with scoring model outputs on a 5-point Likert scale within a dual-axis evaluation framework, constructing a robust, objective, and comprehensive scoring system.

3.4 System Pipeline

3.5 Compute and Feasibility

Our experimental framework leverages a hybrid computing architecture to optimize resource allocation across the generation and evaluation phases.

For the response generation and model training phase, we utilize a local workstation equipped with a single NVIDIA GeForce RTX 4090 GPU (24 GB VRAM). Model weights are loaded in `bf16` precision to maximize inference efficiency. Hardware profiling indicates that during active generation (at 99% GPU utilization), dedicated VRAM usage peaks at 17.1 GB compared to an 8.8 GB idle baseline, leaving a sufficient memory buffer that eliminates the risk of out-of-memory (OOM) errors during long-context generation. Testing four distinct prompt variations across 100 dialogue records each yields per-prompt processing times ranging from 24.8 to 30.5 minutes, corresponding to an average inference speed of 14.89–18.31 seconds per iteration. Generating the complete batch of 400 responses takes under two hours, confirming that complex workflows for the 4B-parameter model are highly feasible and rapidly iterable on consumer-grade high-end hardware.

For the evaluation phase, we deploy our LLM-as-a-Judge pipeline on a Google Colab environment backed by an NVIDIA T4 GPU. Since evaluation relies on querying external API endpoints (GPT-5.4, Gemini 3 Flash, Claude 4.5 Haiku), heavy computation is offloaded to cloud servers while the local T4 instance functions primarily as an orchestration node, handling asynchronous network I/O and API request management. This provides a stable, scalable, and cost-effective solution for our multi-judge scoring system.

4 Evaluation Framework

To evaluate the fine-tuned mental health support model, we adopt the Affective–Cognitive Agreement Framework, which separates the quality of mental health responses into cognitive support and affective alignment. This framework allows us to

assess not only whether a response is informative and safe, but also whether it emotionally aligns with the user’s needs. We implement the evaluation using a multi-agent judge ensemble combined with statistical significance testing.

4.1 Affective–Cognitive Assessment Rubric

We evaluate each generated response along two complementary axes using a 5-point Likert scale.

Cognitive Support Score (CSS) measures the structural and informational quality of the response across four dimensions: Guidance, Informativeness, Relevance, and Safety.

Affective Resonance Score (ARS) measures the emotional alignment between the response and the user’s expressed concerns across three dimensions: Empathy, Helpfulness, and Understanding.

In the current evaluation setup, all seven dimensions contribute equally to the final score. We adopt equal weighting to avoid introducing subjective priors about the relative importance of individual dimensions and to maintain transparency. The final Overall Score is computed as the arithmetic mean of the seven dimension scores.

4.2 LLM-as-a-Judge Ensemble

Automated evaluation of mental health dialogue can be sensitive to judge bias, particularly for subjective dimensions such as empathy. To reduce single-model bias, we use a jury-of-judges ensemble composed of three independent state-of-the-art models from different model families: GPT-series, Gemini-series, and Claude-series. Each judge evaluates responses independently under a temperature of 0.0, ensuring deterministic outputs. We aggregate scores from the three judges by taking their mean, which stabilizes evaluation and reduces variance caused by idiosyncratic scoring behavior.

For benchmarking, we construct a test set of 1,000 previously unseen queries covering 23 clinical conditions including anxiety, depression, and self-harm risk. Using identical system prompts, both the baseline model (Qwen3-4B) and the fine-tuned model generate responses to each query. The evaluation framework scores both outputs across the seven rubric dimensions, and the ensemble provides the final aggregated score. Although automated judges provide scalable evaluation, they may still introduce systematic biases; we therefore measure inter-judge agreement (Section ??) to verify consistency across judge models.

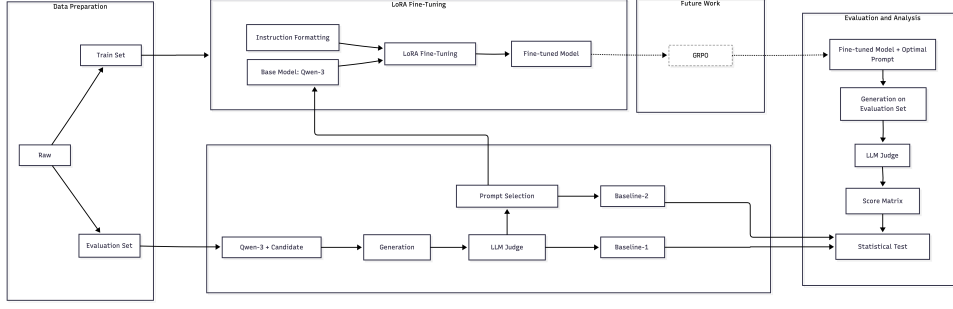


Figure 1: Overview of the proposed strategic alignment pipeline.

4.3 Statistical Measurement and Significance Testing

We measure model improvements using a statistical pipeline that captures performance frequency, improvement magnitude, and statistical significance.

4.3.1 Comparative Performance Metrics

Win Rate (WR) measures how often the fine-tuned model produces a higher score than the baseline, with ties contributing half a win:

$$WR = \frac{\text{Count}(V_{\text{new},j} > V_{\text{old},j}) + 0.5 \times \text{Count}(V_{\text{new},j} = V_{\text{old},j})}{N} \quad (1)$$

Relative Clinical Gain (RCG) measures overall improvement across all samples and evaluation dimensions:

$$RCG = \frac{\sum_{i=1}^{N \times D} (S_{\text{new},i} - S_{\text{old},i})}{\sum_{i=1}^{N \times D} S_{\text{old},i}} \times 100\% \quad (2)$$

4.3.2 Effect Size and Significance Testing

Because Likert-scale ratings are ordinal and often violate normality assumptions, we use non-parametric statistical tests.

Cliff’s Delta (d) measures the effect size between score distributions without assuming a Gaussian distribution:

$$d = \frac{\sum_{i=1}^n \sum_{j=1}^n \text{sign}(x_i - y_j)}{n^2} \quad (3)$$

Wilcoxon Signed-Rank Test (T) evaluates whether observed improvements are statistically significant:

$$T = \sum_{i=1}^N \text{sgn}(X_i) R_i \quad (4)$$

5 Experiments and Progress

5.1 Experimental Setup

All generations use a standardized configuration (same parameters and max output length) and a

fixed system prompt template to ensure fair comparisons across versions. For evaluation, we apply an anchored rubric with an LLM-judge environment, and we log all prompts, model versions, and random seeds to support reproducibility.

5.2 Experiment 1: Baseline Analysis

5.2.1 Baseline

During preliminary experiments with MentalBench-100K, we observed a systematic issue affecting Qwen3-4B generations. Specifically, a subset of Qwen3-4B responses in the dataset are unusually short, partially truncated, or missing, and some entries appear to include explicit “thinking” traces as part of the stored response. This not only compresses the final response length but also contaminates downstream scoring.

To quantify the impact of truncation, we conducted a correlation analysis between response length (output tokens) and evaluation scores in MentalAlign-70k. We found a highly significant, strong positive correlation between the two variables (Pearson’s $r = 0.960$, $p < 0.001$; Spearman’s $\rho = 0.929$, $p < 0.001$). This statistical evidence indicates that longer outputs consistently receive higher rubric scores. Consequently, the observed length artifacts can systematically depress Qwen-3’s measured performance, rendering the original baseline unreliable for fair comparison.

As a consequence, we decided not to use the Qwen3 responses provided in MentalBench-100K as our baseline outputs. We re-established a controlled Qwen3-4B baseline under a standardized generation setup. Concretely, we disabled thinking mode for Qwen3 to ensure that internal reasoning traces do not consume the output budget, thereby preserving a comparable number of tokens for user-facing content under the same max_tokens constraint. While disabling thinking may reduce

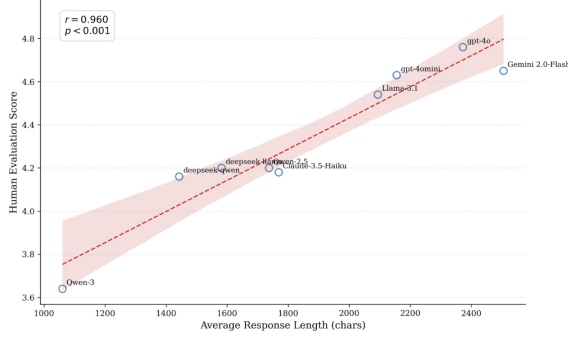


Figure 2: Correlation between response length and human evaluation scores in MentalAlign-70k.

the model’s maximum reasoning expressiveness, it yields a baseline that is more faithful to the evaluation conditions. We then re-evaluated the regenerated Qwen3 outputs using a multi-judge method. The resulting scores constitute our revised Qwen3-4B baseline, which we use in all subsequent experiments and comparisons.

5.2.2 Prompt Engineering

To establish a reliable prompting setup for mental-health response generation, we conducted a prompt engineering study prior to the main experiments. The baseline prompt (v0) was adopted from the prompt-to-model setup used in [Badawi et al. \(2025\)](#), which served as a literature-grounded starting point. Building on this baseline, we designed three additional prompt variants (v1, v2, and v3) to improve different aspects of response quality, with refinements focused on contextual grounding, clinical clarity, practical guidance, and the balance between professional structure and emotional support.

To compare the four prompts under controlled conditions, we sampled 100 instances from MentalBench-100K and used Qwen3-4B to generate responses for each prompt version. All outputs were evaluated using a unified multi-judge framework, and scores were averaged across judges for each dimension. We selected the prompt with the highest overall mean score for subsequent experiments.

As shown in Table 1, v2 achieved the best overall performance (4.347) among all prompt variants. Compared with v0 and v1, it delivered the largest improvements in Guidance and Informativeness while preserving competitive performance on Relevance and Safety. This indicates that v2 offers the best overall balance for generating clinically

grounded and practically useful mental-health responses. Beyond mean score comparisons, we also used a comparative win-rate metric against the baseline (v0). As presented in Table X, v2 achieves the highest win rate (60%, 56 wins / 8 ties out of 100), outperforming v1 (57.5%) and v3 (12%), which further supports selecting v2 as the best-performing prompt variant. Accordingly, we used v2 in all subsequent experiments.

5.3 Experiment 2: Improved Method and Comparative Analysis

We aim to evaluate an improved optimization pipeline that extends the baseline prompt-engineering setup with QLoRA-based supervised fine-tuning (SFT) and RLAIIF via GRPO, followed by rubric-based evaluation using an LLM jury. However, both the training components and the evaluation procedure rely on external LLM APIs for scoring and feedback. As a result, the experiment is cost-sensitive, and we have proceeded cautiously in scaling up runs before completing sufficient engineering optimizations. At the time of writing, this component remains in progress. We will update this section once the full set of runs is completed.

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Version	Guidance	Informativeness	Relevance	Safety	Empathy	Helpfulness	Understanding	Overall
v0	3.533	3.587	4.470	4.670	4.847	4.177	4.540	4.260
v1	3.770	3.783	4.530	4.753	4.727	4.183	4.563	4.330
v2	4.110	4.033	4.537	4.683	4.430	4.180	4.457	4.347
v3	2.683	2.683	4.250	4.523	4.860	4.007	4.410	3.917

Table 1: Prompt variant comparison on 100 sampled instances from MentalBench-100K. Scores are averaged across multi-judge evaluations. Bold indicates the best score per dimension.

Table 2: Win-rate against the baseline prompt (v0) on 100 test cases; v2 performs best (60.0%)

Comparison	Win Rate	Wins	Ties	Total
v1 vs v0	57.50%	55	5	100
v2 vs v0	60.00%	56	8	100
v3 vs v0	12.00%	11	2	100

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